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DataMining Capstone Project
Final Review

Recommendation Analysis Using Library Data

Presented by Section AIML-2

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ABSTRACT

This project aims to develop a Book Recommendation System using Library Data by analyzing both book information and user activity datasets. With the increasing volume of books and users in libraries, recommending relevant books can enhance user experience and improve reading engagement.

• **Book Dataset** – Includes attributes such as User ID, Book ID, Book Title, Genre, Preferred Category, StudentDepartment, Academic Year and Reading Frequency, representing detailed information about available books in the library.

•**Library Borrowing Dataset** – Includes attributes such as Student ID, Book ID, Book Title, Author, Category, Department, UserRating and AcademicYear representing user behavior and interests.

Data preprocessing was performed using Python, which included handling missing values, removing duplicate and noisy data, standardizing formats, validating data ranges, and visualizing data distributions for better understanding.

After cleaning and preprocessing, both datasets were integrated using common attributes such as Book ID and User ID. The final unified dataset was exported to WEKA, where data mining techniques such as classification, clustering, and association rule mining were applied to identify reading patterns and generate personalized book recommendations.

This project demonstrates how data mining techniques can improve library management systems and provide intelligent, user-centered book recommendations.

Data Preprocessing Techniques

Purpose of Data Preprocessing:

- Improve data quality and consistency
- Handle incomplete and noisy data
- Prepare dataset for machine learning algorithms
- Ensured accuracy, consistency, and reliability of the dataset

- **Tools Used:**

  Python

  Pandas

  Matplotlib

- Data preprocessing is a crucial step because machine learning models perform better on clean and structured data.

Handling Missing Values

Step 1: Identifying Missing Values

- Used isnull().sum() to detect missing values
- Checked each column individually

Step 2: Technique Used

1. Imputation (Primary & Category Methods Used)

Applied Mean Imputation & Mode imputation for numerical columns & categorical columns such as:

- User rating
- BorrowDurationDays
- Genre
- ReadingFrequency
- Department

Reason for Using Mean:

- Works well for normally distributed data
- Maintains the overall average of the dataset
- Simple and computationally efficient to implement

Data Preprocessing Techniques

Row Deletion(Conditional Method):

- Removed rows with missing or invalid User ID.
- Used to ensure data accuracy and maintain consistency in the recommendation model.

Reason:

- Ensures meaningful and reliable analysis results.
- Eliminates invalid data and maintains dataset accuracy.

Example:

- If the same User ID and Book ID combination appeared more than once, the row was removed.
- If BorrowDurationDays = –3, the row was deleted because duration cannot be negative.
- If Publication Year = 1800 or 2050, the row was removed as it falls outside a realistic range.

Data Preprocessing Techniques

Noise removal & Range Validation:

To ensure data correctness, range validation was applied:

- User Rating → 1 to 5
- BorrowDurationDays → Greater than 0
- User Age → 5 to 100
- Publication Year → 1900 to Present Year

Why Range Validation ?

- Removes unrealistic values
- Prevents model confusion
- Improves recommendation accuracy
- Maintains data validity

Outliers and inconsistent values were removed to reduce noise in the dataset.

Data Preprocessing Techniques

Data Integration

- Integrated **Book Dataset** and **Library/User Data** using common attributes like **Book ID** and **User ID**
- Created a single unified dataset for analysis

Why Integration?

- To combine book details with user borrowing behavior
- To identify user reading patterns
- To enable personalized book recommendations
- To prepare complete data for applying data mining algorithms

Examples :

- Combined **Book Title, Genre, Author** with **User Rating and Borrow Duration**
- Merged user borrowing history with book information
- Created a final dataset containing both book features and user activity

Data Preprocessing Techniques

Data Transformation & Export

Column Standardization

- Converted **Genre** and **ReadingFrequency** values to lowercase
- Converted **Department** values to uppercase
- Converted required columns into appropriate numeric data types

Data Visualization

- Used **Scatter Plots** to visualize relationships between attributes
- Used **Bar Charts** to analyze categorical data
- Used **Box Plots** to detect outliers

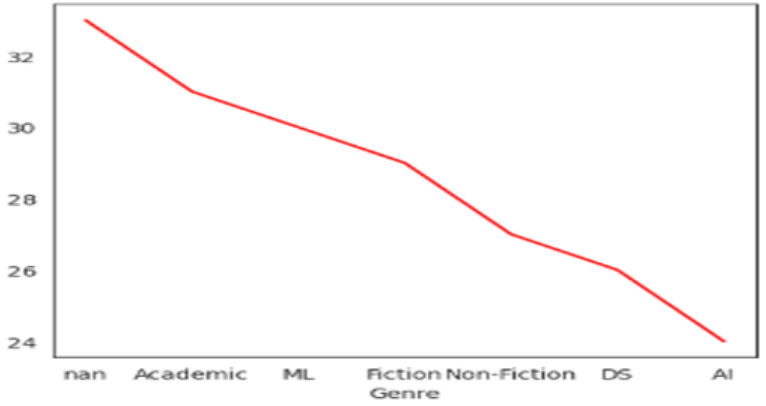
Final Step

- Performed cleaning and transformation on integrated dataset
- Exported the final cleaned dataset as a **CSV file**
- Imported the dataset into **WEKA** to apply classification, clustering, and association algorithms

Data Visualization

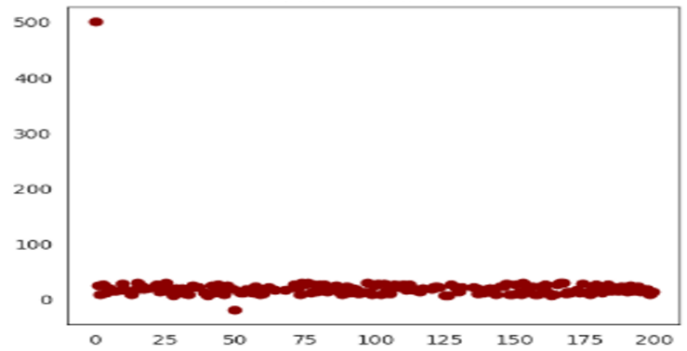
Before Cleaning:

Dataset 1 – Genre Before Cleaning

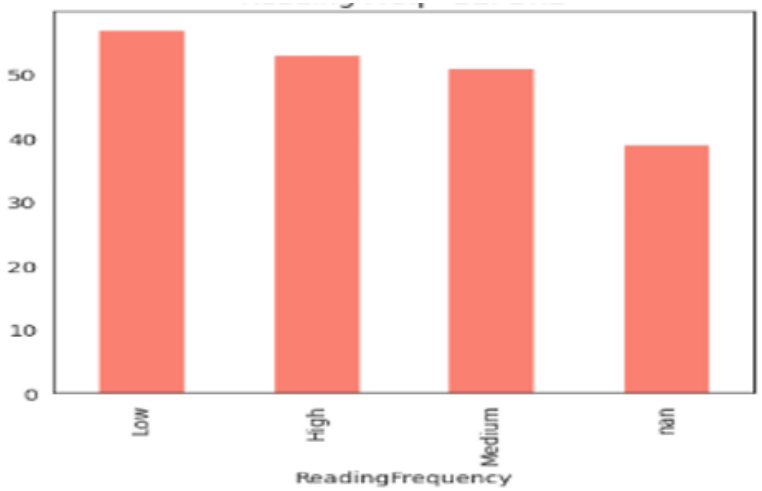


Dataset 2 – : BorrowDurationDays Before cleaning

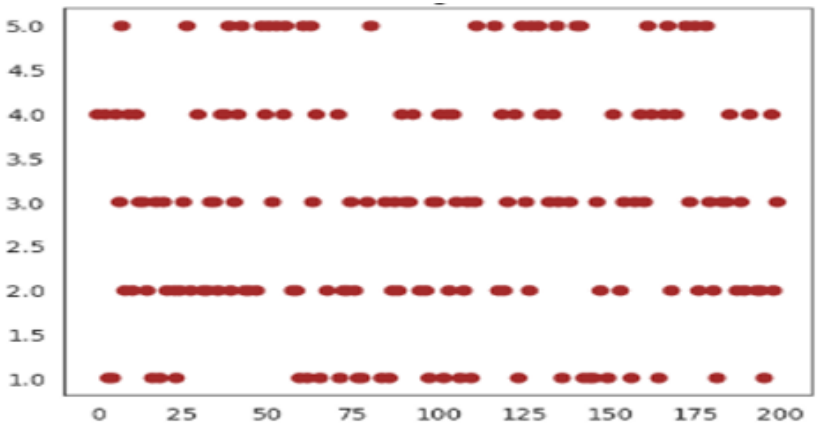
Dataset 2 – Record Index vs BorrowDurationDays



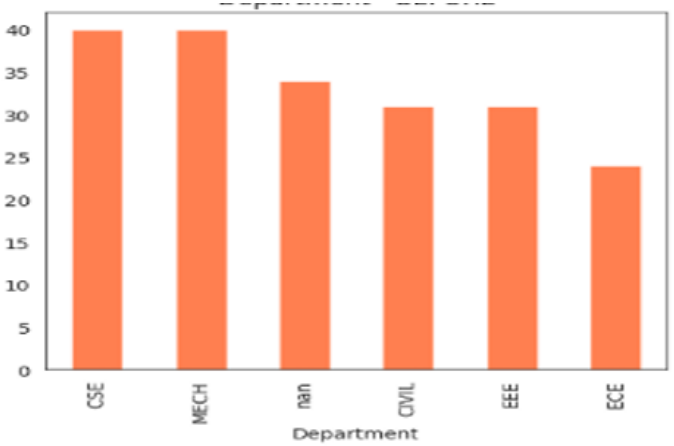
Dataset 2 - ReadingFrequency categories vs Count



Dataset 2 -Record Index vs User Rating



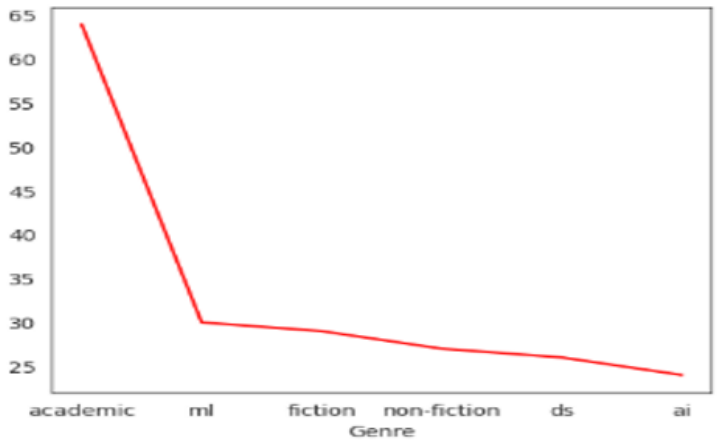
Dataset 2 - Department categories vs Count



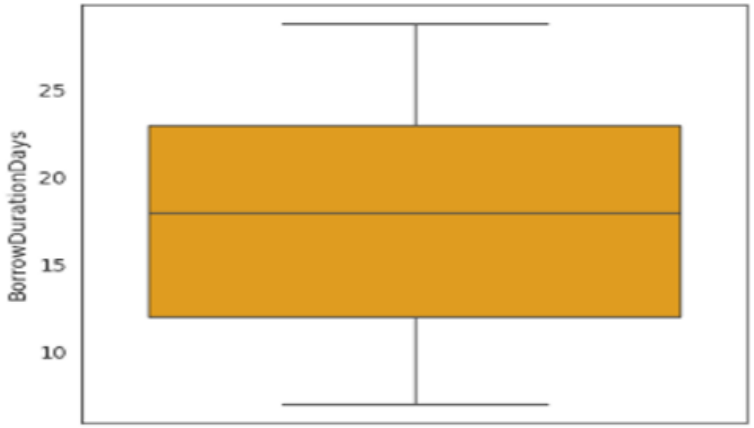
Data Visualization

After Cleaning:

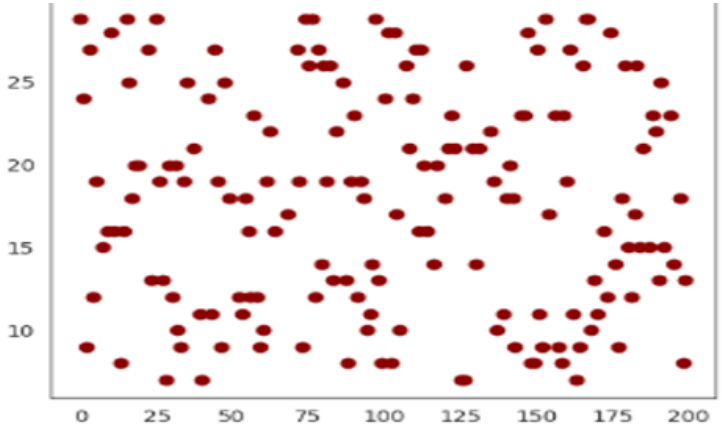
Dataset 1 - Genre categories vs Number of Books



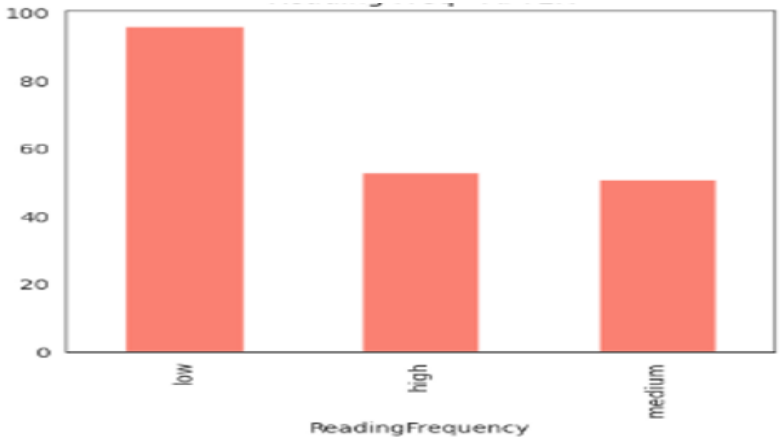
Dataset 2 – BorrowDurationDays after cleeaning



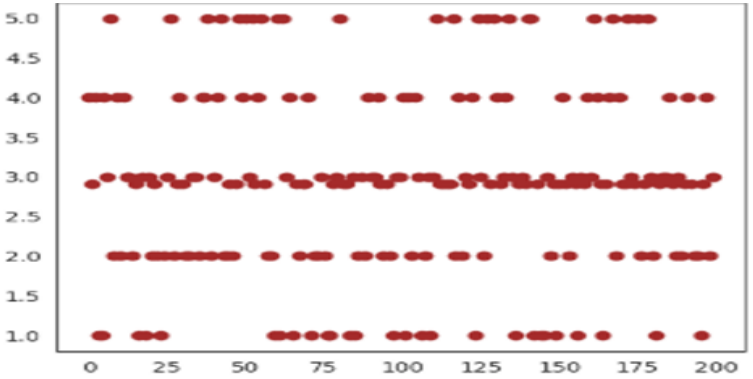
Dataset 2 - Record Index vs BorrowDurationDays



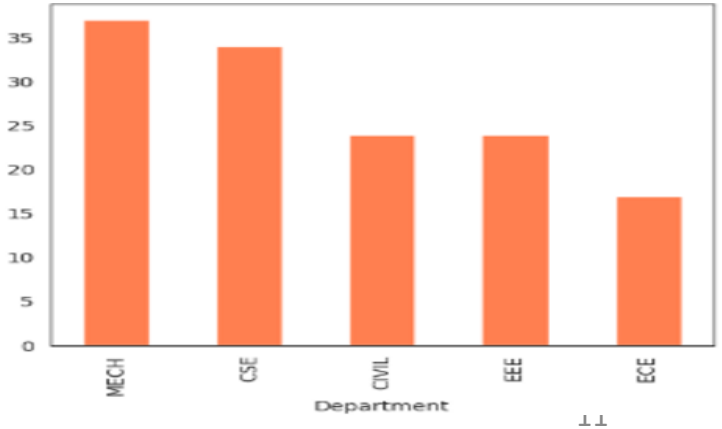
Dataset 2- ReadingFrequency categories vs Count



Dataset 2 - Record Index vs User Rating



Dataset 2 - Department categories vs Count



Data Normalization

Purpose of Normalization
Data normalization is performed to scale numerical attributes into a common range.
It prevents attributes with larger ranges from dominating the model.

Normalization in WEKA
Normalization was applied using the **Preprocess tab** in WEKA.

Steps followed:
Loaded the cleaned dataset into WEKA
Selected **Filter** → **Unsupervised** → **Attribute** → **Normalize**
Applied the filter before running algorithms

Type of Normalization Used
Min-Max Normalization
Scales values between 0 and 1
Formula:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

METHODOLOGY

Classification

- Classification is used to predict whether a user will like a book (Like/Not Like or High Rating/Low Rating) based on book attributes and user behavior.

The following classification algorithms were studied:

1. J48 Algorithm

- J48 is a decision tree algorithm that splits the dataset based on attributes.
- It selects the best attribute using information gain and builds a tree structure.
- It is easy to understand and helps in making predictions effectively.

2. Random Tree

- Random Tree is another decision tree-based algorithm that selects attributes randomly.
- It is faster and helps in reducing overfitting.
- It works well with large datasets and provides good prediction results.

Association Rule Mining

Association Rule Mining is used to discover hidden relationships between attributes in the dataset (using If–Then rules).

It helps identify patterns such as:

High ReadingFrequency + High User Rating → High Interest in a Particular Genre

The following algorithms were studied:

1. Apriori Algorithm

- Identifies frequent item combinations in the dataset
- Works based on minimum support and confidence thresholds
- Generates reliable and meaningful association rules
- Appropriate for small and medium-scale datasets

METHODOLOGY

Clustering

- Clustering is an unsupervised learning technique used to group similar data points.

1. K-Means Algorithm

- K-Means divides data into a fixed number of clusters.
- Each cluster has a centroid, and data points are assigned based on distance from the centroid.

Steps Involved:

- Select number of clusters (K)
- Initialize centroids
- Assign data points to nearest centroid
- Update centroids
- Repeat until convergence

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OUTPUT SCREEN

J48 Algorithm:

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Classifier

Choose: J48 -R -N 3 -Q 1 -M 5

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 10
- Percentage split — % 66

More options...

(Nom) RatingLevel

Start | Stop

Result list (right-click for options)

11:18:30 - trees.J48

Classifier output

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -R -N 3 -Q 1 -M 5
Relation:     final-weka.filters.unsupervised.attribute.Remove-R1-3,5,7-11,13-21,23,25
Instances:    200
Attributes:   5
              Genre
              ReadingFrequency
              AcademicYear_x
              Department
              RatingLevel
Test mode:    evaluate on training data

=== Classifier model (full training set) ===

J48 pruned tree
------------------

Genre = ml
|   AcademicYear_x = 3rd Year: low (11.0/6.0)
|   AcademicYear_x = 1st Year: medium (2.0)
|   AcademicYear_x = 4th Year: low (2.0)
|   AcademicYear_x = 2nd Year: medium (5.0)
Genre = non-fiction: low (24.0/8.0)
Genre = academic
|   Department = EEE: medium (5.0/2.0)
|   Department = MECH: medium (3.0/1.0)
|   Department = CSE: high (7.0/3.0)
|   Department = CIVIL
|   |   AcademicYear_x = 3rd Year: low (0.0)
|   |   AcademicYear_x = 1st Year: high (1.0)
|   |   AcademicYear_x = 4th Year: low (8.0)
|   |   AcademicYear_x = 2nd Year: medium (9.0/3.0)
|   Department = ECE: high (3.0)
```

Status: OK | Log | x 0

OUTPUT SCREEN

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Classifier

Choose: J48 -R -N 3 -Q 1 -M 5

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 10
- Percentage split — % 66

More options...

(Nom) RatingLevel

Start | Stop

Result list (right-click for options)

11:18:30 - trees.J48

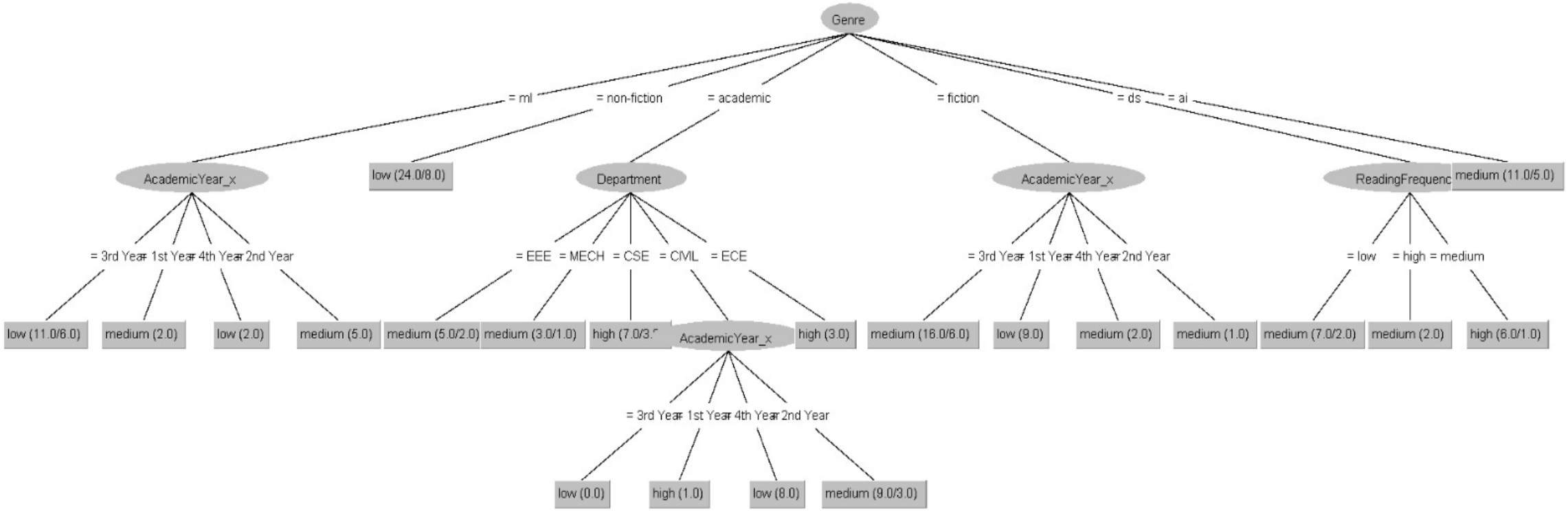
Classifier output

```
Time taken to build model: 0 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances         148               74      %
Incorrectly Classified Instances        52               26      %
Kappa statistic                          0.5786
Mean absolute error                      0.2358
Root mean squared error                  0.3417
Relative absolute error                 56.4948 %
Root relative squared error             74.8332 %
Total Number of Instances              200

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.784    0.151    0.753      0.784    0.768      0.628    0.910     0.843     low
                 0.747    0.248    0.716      0.747    0.731      0.498    0.831     0.792     medium
                 0.629    0.036    0.786      0.629    0.698      0.648    0.929     0.730     high
Weighted Avg.    0.740    0.175    0.742      0.740    0.739      0.573    0.877     0.800

=== Confusion Matrix ===

  a  b  c   <-- classified as
 58 16  0 |  a = low
 17 68  6 |  b = medium
  2 11 22 |  c = high
```

Status: OK

Log | x 0

OUTPUT SCREEN



OUTPUT SCREEN

Random tree:

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Classifier

Choose **RandomTree** -K 0 -M 10.0 -V 0.001 -S 1 -depth 5

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 10
- Percentage split — % 66

More options...

(Nom) RatingLevel

Start | Stop

Result list (right-click for options)

11:30:01 - trees.RandomTree

Classifier output

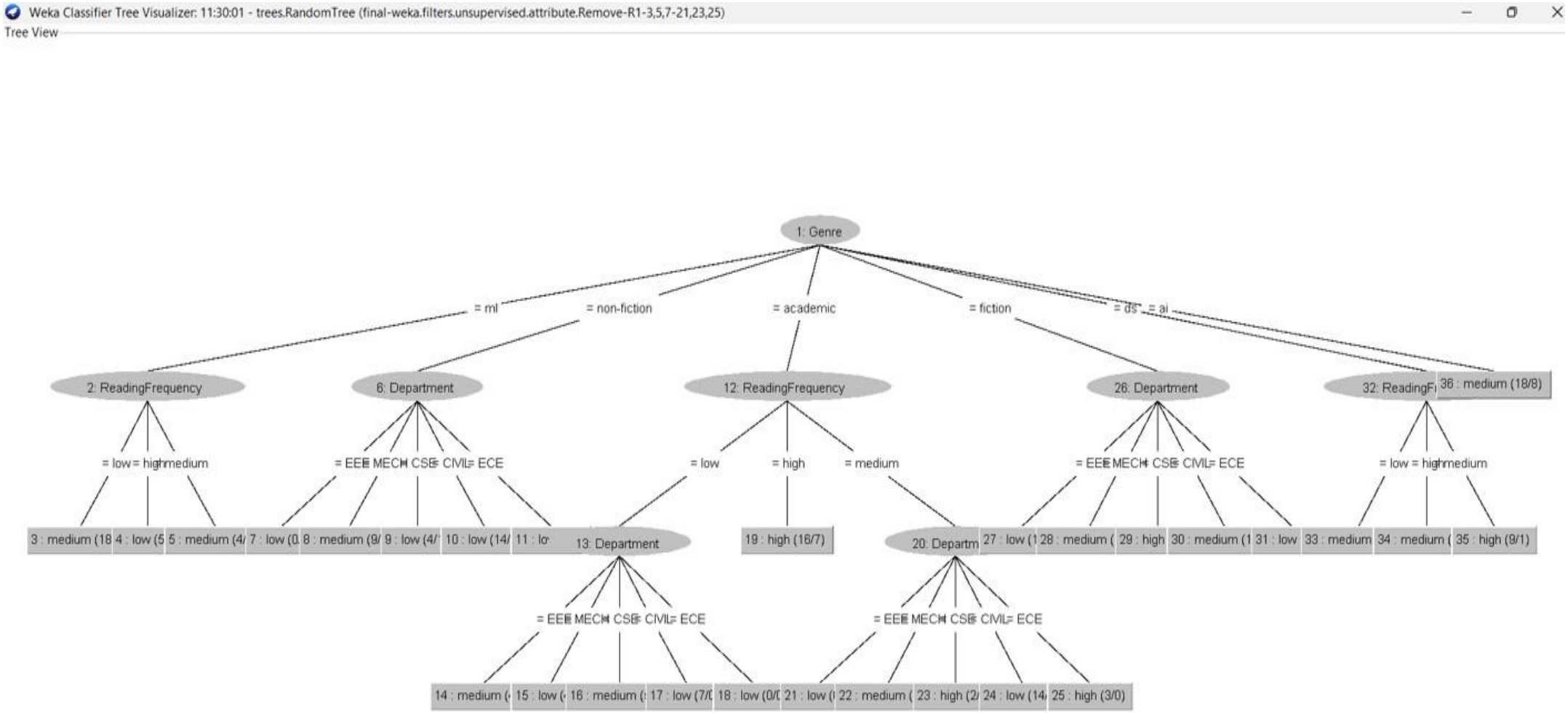
```
Size of the tree : 36
Max depth of tree: 5

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.02 seconds

=== Summary ===

Correctly Classified Instances         142               71      %
Incorrectly Classified Instances        58               29      %
Kappa statistic                          0.532
Mean absolute error                      0.252
Root mean squared error                  0.3549
Relative absolute error                 60.3678 %
Root relative squared error             77.734  %
Total Number of Instances              200

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.649    0.151    0.716      0.649    0.681      0.509    0.851     0.756     low
                 0.758    0.284    0.690      0.758    0.723      0.472    0.820     0.785     medium
                 0.714    0.048    0.758      0.714    0.735      0.682    0.957     0.803     high
Weighted Avg.    0.710    0.194    0.712      0.710    0.709      0.522    0.856     0.777

=== Confusion Matrix ===

  a  b  c   <-- classified as
 48 23  3 |  a = low
 17 69  5 |  b = medium
  2  8 25 |  c = high
```

Status

OK

Log x 0

OUTPUT SCREEN



OUTPUT SCREEN

Apriori Algorithm:

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Associator

Choose **Apriori** -I -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start | Stop

Result list (right-click for ...

11:54:19 - Apriori

Associator output

```
=== Run information ===

Scheme:       weka.associations.Apriori -I -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1
Relation:     final-weka.filters.unsupervised.attribute.Remove-R1-3,5,7-11,13-21,23-weka.filters.unsupervised.attribute.NumericToNominal-Rfirst-last
Instances:    200
Attributes:   6
              Genre
              ReadingFrequency
              AcademicYear_x
              Department
              RatingLevel
              BorrowDurationLevel
=== Associator model (full training set) ===


Apriori
=======

Minimum support: 0.1 (20 instances)
Minimum metric <confidence>: 0.7
Number of cycles performed: 18

Generated sets of large itemsets:

Size of set of large itemsets L(1): 22

Large Itemsets L(1):
Genre=ml 27
Genre=non-fiction 32
Genre=academic 57
Genre=fiction 43
Genre=ds 23
ReadingFrequency=low 81
ReadingFrequency=high 56
ReadingFrequency=medium 63
AcademicYear_x=3rd Year 58
AcademicYear_x=1st Year 43
```

Status

OK

Log | x 0

OUTPUT SCREEN

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Associator

Choose **Apriori** -I -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start | Stop

Result list (right-click for ...

11:54:19 - Apriori

Associator output

```
AcademicYear_x=4th Year RatingLevel=low 26
AcademicYear_x=4th Year RatingLevel=medium 25
AcademicYear_x=4th Year BorrowDurationLevel=medium 32
AcademicYear_x=2nd Year Department=CIVIL 26
AcademicYear_x=2nd Year RatingLevel=medium 25
AcademicYear_x=2nd Year BorrowDurationLevel=medium 25
Department=EEE BorrowDurationLevel=long 25
Department=MECH RatingLevel=medium 25
Department=CIVIL RatingLevel=low 38
Department=CIVIL RatingLevel=medium 33
Department=CIVIL BorrowDurationLevel=long 34
Department=CIVIL BorrowDurationLevel=medium 42
RatingLevel=low BorrowDurationLevel=long 29
RatingLevel=low BorrowDurationLevel=medium 34
RatingLevel=medium BorrowDurationLevel=long 37
RatingLevel=medium BorrowDurationLevel=medium 37

Size of set of large itemsets L(3): 6

Large Itemsets L(3):
Genre=fiction RatingLevel=medium BorrowDurationLevel=long 20
ReadingFrequency=low RatingLevel=medium BorrowDurationLevel=medium 20
ReadingFrequency=medium Department=CIVIL RatingLevel=low 20
ReadingFrequency=medium Department=CIVIL BorrowDurationLevel=medium 28
AcademicYear_x=3rd Year RatingLevel=medium BorrowDurationLevel=long 21
Department=CIVIL RatingLevel=low BorrowDurationLevel=medium 20

Best rules found:

 1. ReadingFrequency=medium RatingLevel=low 23 ==> Department=CIVIL 20    <conf:(0.87)> lift:(2) lev:(0.05) [9] conv:(3.25)
 2. Department=EEE 29 ==> BorrowDurationLevel=long 25    <conf:(0.86)> lift:(2.08) lev:(0.06) [12] conv:(3.39)
 3. Genre=fiction RatingLevel=medium 24 ==> BorrowDurationLevel=long 20    <conf:(0.83)> lift:(2.01) lev:(0.05) [10] conv:(2.81)
 4. Genre=fiction 43 ==> BorrowDurationLevel=long 33    <conf:(0.77)> lift:(1.85) lev:(0.08) [15] conv:(2.29)
 5. AcademicYear_x=3rd Year RatingLevel=medium 28 ==> BorrowDurationLevel=long 21    <conf:(0.75)> lift:(1.81) lev:(0.05) [9] conv:(2.05)
 6. ReadingFrequency=medium BorrowDurationLevel=medium 38 ==> Department=CIVIL 28    <conf:(0.74)> lift:(1.69) lev:(0.06) [11] conv:(1.95)
 7. ReadingFrequency=medium 63 ==> Department=CIVIL 46    <conf:(0.73)> lift:(1.68) lev:(0.09) [18] conv:(1.98)
```

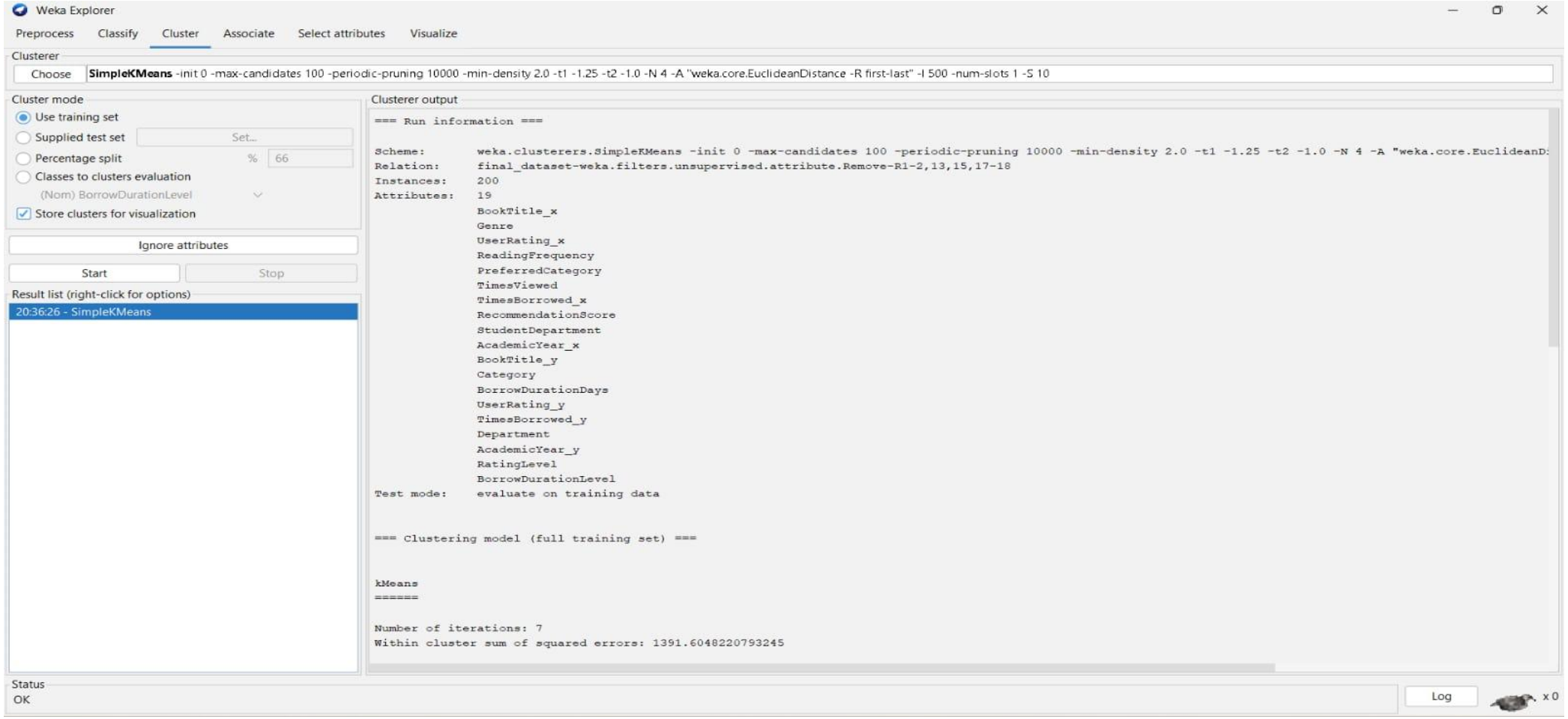
Status

OK

Log | x 0

OUTPUT SCREEN

Simple Kmeans Algorithm:



OUTPUT SCREEN

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 4 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- Use training set
- Supplied test set — Set...
- Percentage split — % 66
- Classes to clusters evaluation
  - (Nom) BorrowDurationLevel
- Store clusters for visualization

Ignore attributes

Start | Stop

Result list (right-click for options)

20:36:26 - SimpleKMeans

Clusterer output

```
Initial starting points (random):

Cluster 0: Book_139,ds,2,low,AI,6,6,3,CSE,'1st Year',Book_151,ML,26,2,4,MECH,'4th Year',low,long
Cluster 1: Book_100,fiction,2,high,DS,11,10,2.59,ECE,'1st Year',Book_168,'Data Science',27,5,4,EEE,'1st Year',low,long
Cluster 2: Book_123,academic,2.904459,medium,AI,9,9,4.04,CIVIL,'2nd Year',Book_100,Electronics,7,1,7,CIVIL,'4th Year',medium,short
Cluster 3: Book_143,academic,2,high,DS,16,3,3.83,EEE,'1st Year',Book_8,Electronics,16,5,7,EEE,'1st Year',low,medium

Missing values globally replaced with mean/mode

Final cluster centroids:
                                   Cluster#
Attribute               Full Data          0           1           2           3
                          (200.0)     (46.0)      (56.0)      (63.0)      (35.0)
==============================================================================
BookTitle_x              Book_159     Book_5    Book_174     Book_38     Book_91
Genre                    academic non-fiction     fiction    academic    academic
UserRating_x                2.923     2.7163      3.1912      3.4278      1.8571
ReadingFrequency              low        low        high      medium         low
PreferredCategory              AI         AI          DS Electronics       Maths
TimesViewed                10.085    11.9783     10.4464      8.9524      9.0571
TimesBorrowed_x             6.275     5.8913      6.4107      5.7302      7.5429
RecommendationScore        2.3933     2.2689      1.9604      2.6429         2.8
StudentDepartment           CIVIL        CSE         ECE       CIVIL         EEE
AcademicYear_x           3rd Year   4th Year    3rd Year    2nd Year    4th Year
BookTitle_y               Book_58   Book_186     Book_58    Book_113     Book_6
Category              Mathematics Mathematics Mathematics          ML          AI
BorrowDurationDays        18.4465     15.413     24.1563     16.9135     16.0571
UserRating_y                    3     2.4565      4.3214           2         3.4
TimesBorrowed_y              4.96     4.3261      4.9643      4.6984      6.2571
Department                  CIVIL       MECH         EEE       CIVIL       CIVIL
AcademicYear_y           4th Year   3rd Year    1st Year    4th Year    2nd Year
RatingLevel                medium     medium      medium      medium         low
BorrowDurationLevel        medium     medium        long      medium      medium
```

Status

OK

Log | x 0

OUTPUT SCREEN

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 4 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- Use training set
- Supplied test set — Set...
- Percentage split — % 66
- Classes to clusters evaluation
  - (Nom) BorrowDurationLevel
- Store clusters for visualization

Ignore attributes

Start | Stop

Result list (right-click for options)

20:36:26 - SimpleKMeans

Clusterer output

```
                          (200.0)      (46.0)      (56.0)      (63.0)      (35.0)
==============================================================================
BookTitle_x               Book_159      Book_5    Book_174     Book_38     Book_91
Genre                     academic non-fiction     fiction    academic    academic
UserRating_x                 2.923      2.7163      3.1912      3.4278      1.8571
ReadingFrequency               low         low        high      medium         low
PreferredCategory               AI          AI          DS Electronics       Maths
TimesViewed                 10.085     11.9783     10.4464      8.9524      9.0571
TimesBorrowed_x              6.275      5.8913      6.4107      5.7302      7.5429
RecommendationScore         2.3933      2.2689      1.9604      2.6429         2.8
StudentDepartment            CIVIL         CSE         ECE       CIVIL         EEE
AcademicYear_x            3rd Year    4th Year    3rd Year    2nd Year    4th Year
BookTitle_y                Book_58    Book_186     Book_58    Book_113      Book_6
Category               Mathematics Mathematics Mathematics          ML          AI
BorrowDurationDays         18.4465      15.413     24.1563     16.9135     16.0571
UserRating_y                     3      2.4565      4.3214           2         3.4
TimesBorrowed_y               4.96      4.3261      4.9643      4.6984      6.2571
Department                   CIVIL        MECH         EEE       CIVIL       CIVIL
AcademicYear_y            4th Year    3rd Year    1st Year    4th Year    2nd Year
RatingLevel                 medium      medium      medium      medium         low
BorrowDurationLevel         medium      medium        long      medium      medium



Time taken to build model (full training data) : 0.03 seconds

=== Model and evaluation on training set ===

Clustered Instances

0       46 ( 23%)
1       56 ( 28%)
2       63 ( 32%)
3       35 ( 18%)
```

Status

OK

Log | x 0

OUTPUT SCREEN

CONCLUSION

- In this project, a book recommendation system was developed using library and user activity datasets.
- Data preprocessing was carried out using Python, including handling missing values, removing duplicate and invalid records, applying range validation, performing standardization, and integrating datasets. This improved the overall quality and consistency of the data.
- Classification algorithms such as Decision Tree and Random Forest were applied in WEKA to predict user interest in books. The model performance was evaluated using accuracy and other performance measures.
- Association rule mining using the Apriori algorithm helped identify meaningful relationships between attributes, such as the connection between reading frequency, genre preference, and user ratings.
- Overall, the project demonstrates how data mining techniques can enhance personalized book recommendations and improve library management systems.

THANK YOU